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The balance between collaborative stability and performance in scientific funding project teams: the role of leader power

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Scientific teams emphasize integrating and generating new knowledge, and team members' constant addition and attrition reflect the dynamic development of team knowledge. However, excessive instability may bring risks and challenges to team development. This study investigates how collaboration dynamics affect team performance in projects funded by the National Natural Science Foundation of China (NSFC). Initially, various indicators were developed to measure project team collaborative stability, performance, and leader power. Then, a hierarchical multiple regression analysis was used to examine the correlation between team collaborative stability and performance, as well as both the direct and moderating effects of team leader power. Results indicate inverted U-shaped relationships between team collaborative stability and performance metrics, peaking at a stability level of approximately 0.3. Findings also suggest leader power can moderate the relationship between collaborative stability and team performance by making the inverted U-shaped relationship curve steeper while shifting its turning point to the left. These findings provide valuable insights and practical guidance for project leaders to effectively manage team collaboration and enhance team performance.

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Introduction

Global competition has heightened the significance of governments leveraging “organized research” to foster original innovations, with government funding playing a pivotal role in encouraging scientific teams to pursue groundbreaking research. Aligning with national policies, the National Natural Science Foundation of China (NSFC) has continuously upgraded and optimized its funding system, strengthening its stable support for innovative scientific teams. The primary goal of NSFC investment is to achieve groundbreaking and innovative outcomes at the frontiers of science, thereby offering a broader perspective for analyzing scientific collaborations. NSFC projects often have an implementation period of three to five years, members actively seek partners with complementary resources and expertise, facilitating dynamic adaptation to the project’s evolving requirements (Zeng et al., 2021). As the project progresses, the team strategically incorporates new members while previous members may disengage as needed. This dynamic evolution of team composition ensures continuous knowledge production and effectively drives project execution (Yao et al., 2025). Given the dynamic and temporal characteristics of NSFC project teams, collaborative stability emerges as an important factor for the success of long-term projects and the advancement of scientific knowledge. The collaboration among team members can significantly influence the efficiency and performance of the project (Abramo et al., 2017; Liu et al., 2022). Insufficient stability is likely to cause deviations in research directions and misallocation of resources, leading to delayed outcomes or a decline in quality (Cooper et al., 2021). For example, in interdisciplinary projects, frequent changes in team members can disrupt the knowledge coordination system and impede the in-depth progress of research. Therefore, an in-depth exploration of the collaborative stability of NSFC project teams is of great practical significance for optimizing the allocation of national scientific research resources and enhancing innovation efficiency.

In ecosystems, stability refers to the ability to resist external disturbances and recover to its original state (Grimm & Wissel, 1997), serving as a critical characteristic of both natural and social processes. Similarly, stability in social networks relies on the effective functioning and sustainable development of cooperative relationships (Guo et al., 2021). Teams, as complex and adaptive systems, evolve over time, where stability signifies enduring partnerships crucial for sustained innovation in scientific contexts (McGrath et al., 2014). While stability can improve performance through shared routines and trust (Kumar & Zaheer, 2019), low-stability collaborations may also yield high-impact results due to personnel mobility (Bu et al., 2018; He et al., 2021). Studies have also observed curvilinear relationships between team stability and performance (Liu et al., 2022; Slotegraaf & Atuahene-Gima, 2011). Despite these insights, previous research has overlooked the measurement of stability in scientific funding project teams, and the relationship between team collaborative stability (TCS) and performance in NSFC project teams remains unclear.

Compared with other research teams, NSFC project teams exhibit distinctive characteristics: strictly defined project cycles, cross-institutional collaboration among members, and a focus on high-impact scientific challenges (Conroy, 2024). The triple constraints of NSFC teams create a distinct stability-performance dynamic. Specifically, rigid timeline pressure necessitates rapid knowledge integration, yet cross-institutional collaboration weakens formal control mechanisms, while national strategic mandates demand breakthrough outputs within limited cycles. To achieve efficiency under these constraints, NSFC project teams must strategically adjust their stability strategies to achieve sustained performance improvements. Consequently, these inherent

characteristics limit the direct applicability of existing theoretical frameworks regarding team stability-performance relationships to NSFC project team management practices. Conducting research on this group can not only yield more targeted theoretical results but also provide direct guidance for the management practices of scientific research projects. Thus, this study proposes RQ1: How does collaborative stability affect team performance in the context of NSFC-funded projects?

Recent research emphasizes the pivotal role of research leader power in fostering collaboration, particularly its ability to integrate resources to initiate, advance, and sustain projects, leading to higher productivity and impact (He et al., 2021; Klavans & Boyack, 2010). Leader power is often represented by authorship order in publications, with the first and corresponding authors typically leading collaborations (Wang & Wang, 2017). Social network analysis offers insights into team leader power, with leaders scoring high on centrality indicators and facilitating information and resource access (He & Hu, 2021; Risselada et al., 2016). Leader power, derived from both formal position and informal influence within the team structure, not only reduces intrateam conflict (Wakefield et al., 2008) but also positively impacts team performance (Robert & You, 2018). In NSFC projects, the principal investigator (PI) serves as the team leader throughout the project lifecycle. The temporally constrained nature of NSFC projects necessitates short-term, high-intensity collaboration, among which the leader power becomes critical for sustaining team cohesion. Simultaneously, the imperative for breakthrough outcomes demands that leaders leverage expert power to guide research direction through scholarly authority. Furthermore, the cross-institutional and interdisciplinary composition of these teams requires leaders to efficiently coordinate internal scientific research tasks and integrate and optimize external cooperation resources simultaneously. These constraints significantly increase the complexity of leader power and trigger a power reconfiguration phenomenon distinct from conventional organizational settings. The maintenance of NSFC team collaborative stability requires the strategic deployment of multi-dimensional power by leaders.

Given the uniqueness of NSFC projects and the complexity of leader power, existing literature lacks in-depth exploration of the uniqueness of the leader power of scientific funding project teams and the internal relationships between leader power, collaborative stability, and performance. This study systematically explores the differential regulatory paths and action mechanisms of leader power on team performance at different levels of collaborative stability, filling the theoretical void in the context of scientific funding projects and providing new academic perspectives for leader power research. In terms of practical value, the results will provide quantitative standards and practical guidelines for the selection and cultivation of leaders of scientific funding project teams, helping scientific research management departments design professional training systems. Moreover, the relevant experience can be extended to other scientific teams and knowledge-intensive organizations, providing a universal reference framework for the leader power development of innovation-driven teams. Therefore, this study proposes RQ2: To what extent does leader power influence the relationship between collaborative stability and team performance within scientific funding projects?

To address these research questions, we collected public information on completed NSFC general projects from 2011 to 2021 in the Management Science field. Considering the dynamic nature of NSFC teams, it is important to understand how collaborative stability within these teams correlates with positive outcomes, including productivity and plan adherence. These are

important performance indicators for projects because delivering more valuable outputs and doing so with efficiency is more important than ever in today's hypercompetitive environment (Conroy, 2024). In addition, we examined the direct and moderating role of leader power to investigate how team leaders who are responsible for projects should manage the team collaboration process to improve team performance. Our findings have significant implications for understanding the scientific collaboration process and facilitating informed decision-making to optimize team performance.

Literature review and hypotheses development

Effect of collaborative stability on team performance. Previous studies have highlighted the crucial role of the collaborative process in team performance (Crawford & Lepine, 2013; Yao et al., 2025), mainly focused on how teams undergo structural changes to achieve higher performance and adaptability (Beersma et al., 2009). The Structural Adaptation Theory (SAT) suggests teams seek adaptable structures while maintaining stability, but historical factors can disrupt coordination and lead to suboptimal performance during structural changes. Long-term partnerships within teams are crucial for effective cooperation and improved performance (Huckman et al., 2009; Kim et al., 2022), as stable structures enhance familiarity, trust, and coordination among members (Johnson et al., 2006; Robert et al., 2009; Stuart, 2017). Empirical studies show repeated cooperation outperforms new cooperation in trust, commitment, reciprocity, and norm establishment (Liu et al., 2022), reducing errors and improving plan adherence (Huckman & Staats, 2011). Medium to high collaborative stability in scientific teams often leads to optimal performance (Bu et al., 2018), facilitating effective plan implementation and fostering mutual supervision and steady progress.

Transactive Memory System Theory (TMS) suggests that team members function as external cognitive aids for each other, leveraging collective knowledge through a shared understanding of individual expertise (Wegner, 1995). According to TMS, previous cooperative experiences enable team members to recognize each other's expertise, leading to a more efficient division of labor and task coordination (Wageman, 2001). Empirical studies suggest that the TMS can facilitate team innovation performance by enabling teams to utilize their collective knowledge better, thereby achieving higher performance than those without such a system (He & Hu, 2021; Wang et al., 2018). TMS accelerates decision-making by ensuring rapid and coordinated access to high-quality, task-relevant knowledge. Team interaction processes, particularly those involving established cooperative relationships, are pivotal in TMS development (Huang & Chen, 2018). Members with prior cooperative relationships can more accurately identify and share relevant knowledge within the team, utilizing it to achieve desired outcomes. Therefore, well-acquainted team members are more likely to develop a TMS.

While stable collaboration helps teams save time and reduce uncertainty, it also restricts members' opportunities to acquire useful knowledge from a broader range of partners, potentially hindering innovative thinking and performance in knowledge-intensive tasks (Bertolotti et al., 2015). Scientific teams focus on the complementarity and fusion of knowledge. Due to self-reinforcing mechanisms and inertia, reliance on fixed partnerships can weaken the initial complementarity of diverse knowledge domains over time, leading to knowledge homogeneity (Lewis et al., 2007). Thus in creativity-demanding tasks, instability can be advantageous, often fostering innovation (Iino et al., 2021). Scientific innovation often emerges from the recombination of existing knowledge and ideas, which continuous collaboration with new members stimulates. Cognitive diversity enhances

complex problem-solving by integrating different resources (van Knippenberg & Mell, 2016). Member mobility increases opportunities for external knowledge acquisition and recombination, which is vital for avoiding stagnation and stimulating scientific knowledge generation (Fernández-Zubieta et al., 2016). Scientific funding project teams are inherently flexible, allowing a dynamic assembly of expertise to address project needs, disbanding upon completion (Heath et al., 2011).

The incorporation of new partners injects new vitality and value into the team, but it also introduces risks and challenges to team development. Frequent mobility can exacerbate internal conflicts. Collaborating with diverse authors consumes more time and resources and is more likely to encounter communication, academic, and professional barriers, potentially affecting the accuracy of final results (Bu et al., 2018; Xu et al., 2015). Additionally, the addition of new partners can challenge team leader power, increasing risks and uncertainties of team development (Young-Hyman & Kleinbaum, 2020). Therefore, changes in team composition can be disruptive because the exit of members can disrupt coordination and work capabilities. New members need time to adapt to established processes and norms before they can contribute effectively. Essentially, engaging in social activities with new members may lead to process losses (Stuart, 2017).

This study focuses on long-term, dynamically evolving NSFC project teams. Building on the in-depth analysis of the previously discussed mechanisms, it aims to explore the intricate relationship between collaborative stability and performance through two internal mechanisms, proposing corresponding theoretical hypotheses. First, the continuous incorporation of new team members can lead to the reorganization of professional ideas, thereby fostering innovation within the team. Conversely, increased stability and fixed member composition may lead to knowledge homogenization, hindering performance. Second, high member mobility results in frequent changes in collaboration partners, resulting in significant conflicts and tensions before trust is established. As stability increases, trust deepens, and common norms and coordination mechanisms are gradually established, which enhance team performance.

These two mechanisms simultaneously influence the relationship between team stability and performance, as illustrated in Fig. 1. When the effects of these mechanisms are combined, an inverted U-shaped relationship between the variables can be observed (Haans et al., 2016). Initially, team instability is positively correlated with performance, as different internal teams are formed to match specific tasks. Additionally, team member mobility allows for the acquisition and recombination of new external knowledge, enhancing team performance. However, excessive instability hinders knowledge integration and stable team development, leading to a decline in team performance. This indicates that while high mobility can benefit teams, but excessive instability escalates internal conflicts, eventually overshadowing linear growth. Consequently, we anticipate that scientific teams with extreme stability (either very high or very low) will exhibit lower performance than those with moderate stability. Based on this analysis, we propose *Hypothesis 1*.

H1. There is an inverted U-shaped relationship between collaborative stability and performance in NSFC project teams. As collaborative stability increases, team performance initially improves; however, beyond a certain threshold, further increases in stability lead to a decline in team performance.

The impact of leader power. Existing research increasingly conceptualizes power as a function of network position rather than merely formal authority (He & Hu, 2021; Risselada et al.,

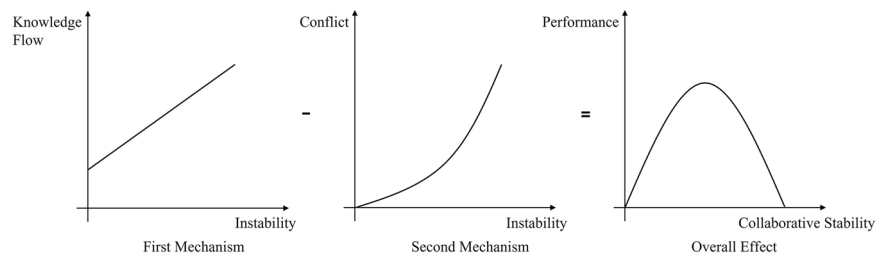


Fig. 1 The relationship between collaborative stability and team performance.

2016; Robert & You, 2018). From this structural perspective, power emanates from an actor's ability to control resource flows by occupying privileged positions within networks (Burt, 1992; Maner & Mead, 2010). Power is widely regarded as the extent of control over valuable resources (Hays et al., 2022). In NSFC project teams, the PI holds the highest status but does not necessarily possess the greatest power. Leader power can significantly enhance team performance by leveraging resources and guiding team dynamics beyond formal organizational boundaries. In knowledge-intensive projects, team leaders are responsible for directing activities, setting research agendas, and contributing to project tasks to expedite progress (O'Kane et al., 2024). Hence, leader power is a critical factor influencing team performance (Rahmani et al., 2018). Although highly self-organized teams exist, virtual teams often perform better under the guidance of a leader (Hertel et al., 2005). Empirical studies have demonstrated that stronger leader power during task execution is able to maximize performance (Kayworth & Leidner, 2002; Lovelace et al., 2001; Wakefield et al., 2008).

Effective leader power hinges on the direct interaction between the leader and team members throughout the team development and performance management processes (Johnson et al., 2015). We model the interaction between team leaders and members using a network representation, assessing the effectiveness of the leader power through the structural hole index of the project PI in the collaboration network. Structural hole theory posits that nodes occupying structure holes possess notable advantages, controlling the flow of resources (Burt, 2004). The information and resources exchanged through these structural holes are highly diverse, providing competitive advantages to these nodes. Researchers situated at structural holes can integrate diverse resources and access non-redundant information from multiple sources, thereby acquiring more advantageous knowledge resources. Furthermore, they can govern the crucial pathways of knowledge flow, ultimately gaining control over resource allocation within the network. As structural hole theory has evolved, some studies have confirmed that occupying structural holes in social networks enhances productivity, creativity, and performance (Wang et al., 2014; Wang et al., 2023). For NSFC project teams, the project PI is the leader with the highest status in the team. When the PI occupies the structure holes in the collaboration network, we hold that the PI has fully exerted leader power, which not only encourages more efficient collaboration among team members through resource allocation but also governs work standards and plan execution. Based on this, we propose *Hypothesis 2*.

H2. The stronger leader power within the NSFC project teams, the higher team performance will be achieved.

A U-shaped relationship can be moderated in two ways. First, the moderator can alter the steepness of the curve, either flattening or steepening it. Second, the moderator can shift the turning point of the relationship between the independent and dependent variables to the left or right. According to the mechanisms described in Fig. 1, team stability and performance

exhibit an inverted U-shaped relationship. When team stability is low, conflicts among members increase. The stronger leader power of the project PI can facilitate the integration of new team members with minimal adjustment, thereby mitigating the negative impacts of these conflicts. Besides, a leader's effective control over resources reduces knowledge loss and knowledge hiding, thus enhancing team performance. Conversely, frequent communication and long-term collaboration establish internal norms in highly stable teams. Severe power differentiation can prevent other members from being empowered or sharing resources. Additionally, the leader's control may restrict communication among members, exacerbating knowledge hiding and hindering the generation of new knowledge, ultimately harming team performance (Hays et al., 2022). Crucially, structural leader power exhibits paradigmatic congruence with our network-based TCS metric. They both operate at relational-structural levels through network mechanisms.

In summary, team leader power affects the inverted U-shaped relationship between team stability and performance in two distinct ways: First, leader power alters the shape of the relationship curve. When team membership is in flux, an effective leader can enhance member familiarity and internal coordination, thereby reducing conflicts. Thus, leader power can modulate the conflict curve in the second mechanism, meaning that effective leader power reduces internal conflicts compared to teams with weaker leader power. Consequently, the inverted U-shaped relationship between team stability and performance becomes steeper. Second, leader power shifts the turning point of the relationship curve. In highly stable teams, leader power can impede knowledge flow, altering the slope of the knowledge flow relationship in the first mechanism. This shift causes the turning point of the inverted U-shaped relationship between team stability and performance to move to the left. The moderating effects of leader power are illustrated in Fig. 2. Based on this analysis, we propose *Hypothesis 3*.

H3. Increasing leader power will steepen the inverted U-shaped relationship between collaborative stability and team performance and shift the turning point of the relationship to the left.

According to the above research hypotheses, the research model framework is constructed as shown in Fig. 3.

Methodology

Data source and sample selection. The National Natural Science Foundation of China (NSFC), established in 1986 and entrusted by the National Ministry of Finance, serves as the primary funding agency for fundamental disciplines in China. NSFC general projects allow researchers to select topics within the funding scope and conduct innovative scientific research in the form of teams, covering a wide range of disciplines and accounting for more than 60% of the total funding.

We crawled the data from the Big Data Knowledge Management Service Portal operated by NSFC (accessible at <https://kd.nsfc.cn/finalProjectInit>). This comprehensive platform

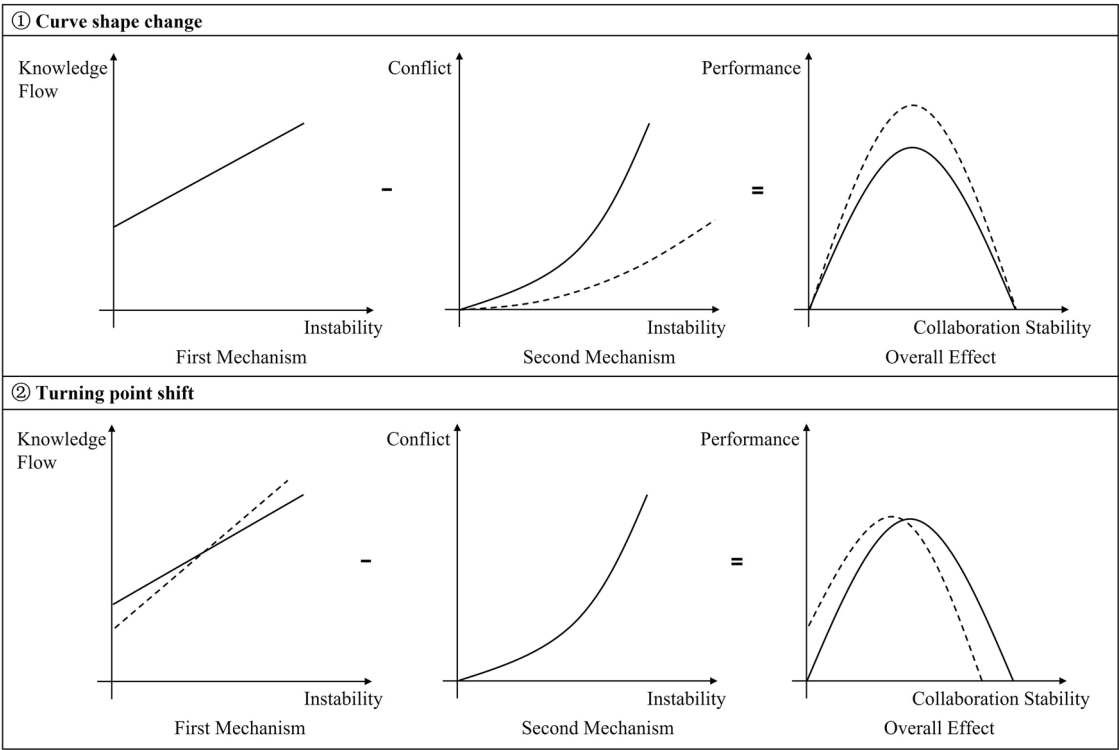


Fig. 2 The moderating effect of team leader power on the inverted U-shaped relationship between collaborative stability and performance.

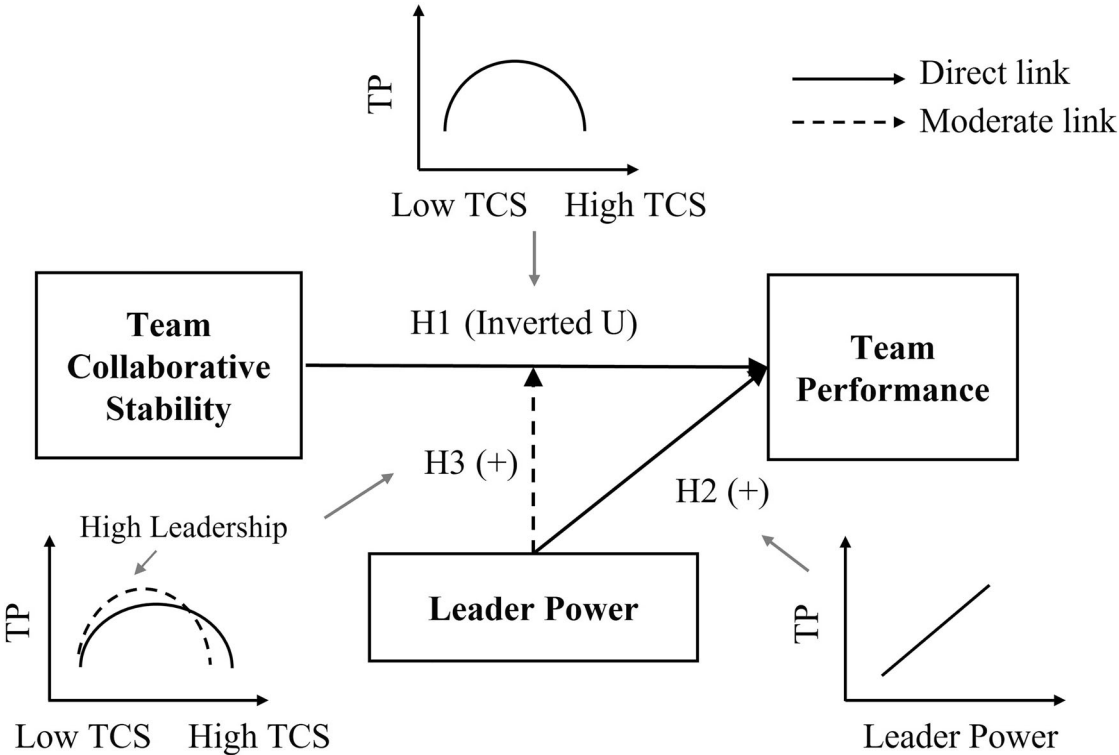


Fig. 3 A model of collaborative stability and performance in NSFC project teams.

caters to research management departments, individual researchers, and the public, offering vital services such as the retrieval and in-depth statistical analysis of funded projects, researchers, and outcomes. Our study collected all successfully concluded general projects within the Management Science field from 2012 to 2018. The dataset includes project information (titles, abstracts,

keywords) and output details (titles, author information, publication dates). Our inclusion criteria for project outputs in the dataset are strictly based on formal grant acknowledgment. Publications with multiple grant numbers are automatically linked to all corresponding projects in our database. Team members in this study are defined as all contributors to project-

related publications. Among these, there are a total of 4,957 projects with a four-year execution period, collectively generating 92,003 outputs.

Operationalization of the variables

Dependent variables

Total productivity: Total productivity is quantified by the total number of team research outputs during project execution, including journal papers, conference papers, patents, monographs, and awards.

Academic productivity: Academic productivity is quantified by the total number of journal and conference publications during project execution.

High-impact productivity: The high-impact productivity is assessed based on the number of high-impact publications, following the measurement framework established by Yao et al. (2025). English high-impact journals are defined as those meeting either of the following criteria: (1) classification as Tier 1 journals in the Chinese Academy of Sciences (CAS) three-year impact factor rankings (top 5%), or (2) inclusion in the University of Texas at Dallas' (UTD24) list of premier academic journals. Our dual-criterion framework not only retains disciplinary breadth through the CAS Tier 1 classification, but also ensures methodological rigor via UTD24's field-specific prestige indicators. Chinese high-impact journals follow the list of important journals identified by the NSFC. High-impact conferences are identified according to Nanjing University's Catalog of Academic Conferences.

Plan adherence: A notable challenge in the execution of NSFC projects is the discrepancy between the outputs and the anticipated research content. We measure plan adherence by calculating the content consistency between the titles of project outcomes and the abstracts outlined in the application.

A parallel comparison approach is employed to measure content consistency between abstracts and outputs of NSFC projects in both Chinese and English. For Chinese texts, the pre-trained BERT-base-Chinese model is used; for English texts, the pre-trained SciBERT model (Reimers & Gurevych, 2019) is utilized. Sentence-BERT (Beltagy et al., 2019), which derives fixed-sized sentence embeddings through a pooling operation on BERT outputs, is selected to quantify content similarity. The cosine similarity between abstract embeddings and output embeddings is then computed, providing a direct measure of similarity. The formula for this calculation is as follows:

$$\text{Sim}(\text{abstract}, \text{output}) = \text{Cos}(x_1, x_2) = \frac{x_1 x_2}{||x_1|| * ||x_2||} \quad (1)$$

x_1 denotes the vector representation of the project abstract, x_2 denotes the vector representation of the project publication titles, $\text{Sim}(\text{abstract}, \text{output})$ is the cosine similarity between the project abstract and the publication titles.

Due to discrepancies in the pre-trained BERT models used for English and Chinese text in this study, indicators were normalized separately for Chinese and English text, with their arithmetic mean serving as the adherence score. The normalization process is articulated as follows in the equation:

$$\text{Normalized}_{\text{score}} = \frac{\text{Score} - \text{Score}_{\min}}{\text{Score}_{\max} - \text{Score}_{\min}} \quad (2)$$

$\text{Normalized}_{\text{score}}$ denotes the normalized score normalization, with values falling within the range of [0,1]. Score represents the score of the current project, $\text{Score}_{\min}$ is the minimum score within the dataset (expect 0), $\text{Score}_{\max}$ represents the maximum score in the

dataset. For projects exclusively producing English or Chinese outputs, the final normalized score directly corresponds to the respective Chinese or English similarity score.

Independent variable. Previous research has proposed metrics to quantify collaborative stability (Bu et al., 2018; Kumar & Zaheer, 2019), but these approaches falter in capturing the fluidity of newly isolated nodes prevalent in real-world project collaborations. In this study, we draw from previous notations of stability and mathematically define a new indicator of project team collaborative stability (TCS) based on the addition and attrition of nodes.

The TCS is calculated using the following formula:

$$TCS_i = 1 - \frac{\sum_{t=1}^y \frac{N_{in} + N_{out}}{N_t}}{y} \quad (3)$$

N_{in} represents the number of new nodes added to the network during time period t , N_{out} represents the number of nodes that exited the network during time period t , and N_t is the total number of unique nodes in the network during time period t . It holds that $N_{in}, N_{out} \geq 0$ and $N_t \geq 1$. To calculate the TCS metric, first we constructed project collaboration networks for each year where nodes represent researchers who contributed to project outputs, and edges denote collaborative relationships. Then, for consecutive years t and $t+1$, we identified emergent nodes (researchers newly participating in year $t+1$) and attributed nodes (researchers active in year t but absent in year $t+1$). Finally, TCS was calculated for each team i using formula (3).

To better illustrate the measurement, we constructed subgraphs of scientific team collaborations for each year. This visualization captures the evolution of both high-stability and low-stability team collaborations. TCS focuses exclusively on collaboration patterns during the project lifecycle. Pre-project collaborations are not considered as they lack the institutional mandate and shared objectives inherent to funded projects. Specifically, if a team's internal researchers maintain stable collaborative relationships and consistently produce collaborative outcomes, the team is considered to have high stability, as shown in Fig. 4(a). Since $N_{in} = N_{out} = 0$, $TCS_a = 1$. Conversely, teams exhibiting low stability are characterized by a persistent inflow of new members and an outflow of previous members throughout the project lifecycle, as visually demonstrated in Fig. 4(b). Following the calculation method and formula described above, $TCS_b = 1 - (13/14 + 8/10 + 1)/3 = 0.09$.

Moderating variable. Constraint is a key metric for calculating structural holes. The more structural holes a participant can control, the less constrained it is (Wang et al., 2023). Accordingly, the leader power of node n is calculated as follows:

$$\text{leaderpower}_n = 2 - C_{nm} = 2 - \sum_m \left(p_{nm} + \sum_q p_{nq} p_{qm} \right)^2 \quad (4)$$

p_{nm} represents the proportion of the relation power between node n and m , relative to all relationships of node n . The same goes for p_{nq} and p_{qm} .

Control variables. To eliminate potential alternative explanations, six additional confounding variables were selected: team size ($size_i$), funding amount ($fund_i$), level of host institution ($level_i$), number of regions ($region_i$), number of institutions ($institution_i$), and number of postgraduates ($student_i$). The definitions are as follows:

- (1) Team Size: The number of members included in the team.
- (2) Funding Amount: Amount of funding received from NSFC.
- (3) Level of Host Institution: This variable is a dummy variable and is equal to 1 if the host institution belongs to Project 985, and 0 otherwise.

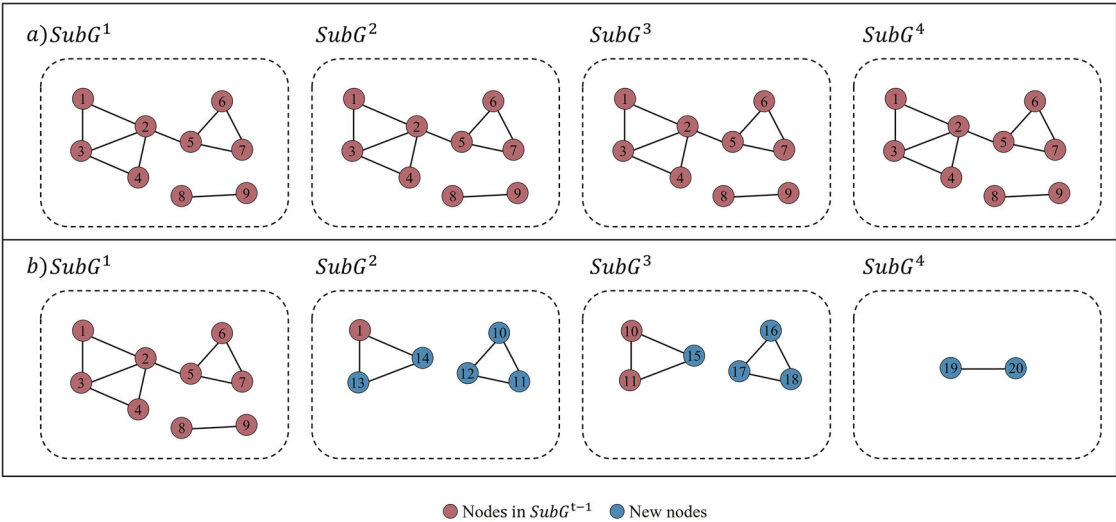


Fig. 4 Examples of project teams with a) high stability and b) low stability.

Table 1 Results of descriptive statistical analysis.						
Variable Type	Variable	Observation	Mean	Std	Min	Max
Dependent	<i>total_prod_i</i>	4957	20.972	15.795	1	182
	<i>academic_prod_i</i>	4957	18.812	14.101	0	175
	<i>high_impact_prod_i</i>	4957	2.702	3.657	0	37
	<i>adherence_i</i>	4957	0.844	0.087	0.09	0.99
Predictor	<i>stability_i</i>	4957	0.243	0.132	0	1
	<i>leaderpower_n</i>	4957	1.750	0.243	0	2
	<i>size_i</i>	4957	8.798	1.364	3	10
Control	<i>fund_i</i>	4957	50.836	6.103	18	69
	<i>level_i</i>	4957	0.526	0.499	0	1
	<i>region_i</i>	4957	2.116	1.045	1	8
	<i>institution_i</i>	4957	3.040	1.435	1	10
	<i>student_i</i>	4957	3.865	1.740	0	9

- (4) Regions: The number of regions included in the team.
- (5) Institutions: The number of institutions included in the team.
- (6) Postgraduate Students: The number of students included in the team.

Data analysis strategy. A negative binomial regression model is utilized for productivity and impact, whereas an OLS linear regression model is employed for plan adherence. Hierarchical multiple regression analysis is used to examine both linear and quadratic relationships between team collaborative stability and team performance, as well as the interaction effect of leader power and stability on team performance. To assess the effects accurately, independent variables are mean-centered to prevent multicollinearity. The predictors are then sequentially entered into the regression equation to test the hypothesized interactions between collaborative stability and leader power and their effect on team performance, following the steps taken by Bertolotti et al. (2015).

In the first step, control variables such as team size, funding amount, host institution level, the number of regions, institutions and students are included to manage potential confounding effects. To control the time-invariant and discipline-invariant factors, fixed effects regarding application year (Y_t) and disciplines (C_i) are also included. The linear term of team stability ($stability_i$) is added to detect the linear relationship between stability and performance. In step 2, the square term of

team stability ($stability_i^2$) is introduced to examine the quadratic main effect of stability on performance. The third step incorporates the leader’s power ($leaderpower_n$) as an explanatory variable to observe its direct impact on team performance. In step 4, the interaction term between team stability and leader power ($leaderpower_n \times stability_i$), as well as the interaction term between leader power and the square of stability ($leaderpower_n \times stability_i^2$), are added to investigate the curvilinear relationship between team stability and performance moderated by leader power.

$$Performance_i = \beta_0 + \beta_1 stability_i + \beta_2 stability_i^2 + \beta_3 leaderpower_n + \beta_4 leaderpower_n \times stability_i + \beta_5 leaderpower_n \times stability_i^2 + \Sigma \beta_6 control_i + Y_t + C_i + \varepsilon_i$$

(5)

Analysis and results
Descriptive statistics. Table 1 reports the results of descriptive statistics for all variables. Before conducting the regression analysis, a variance inflation factor (VIF) analysis was performed to check for multicollinearity effects among the variables in the model. The results indicated that all explanatory variables had VIF values below 3. Therefore, multicollinearity is not a concern in the regression model estimation.

Table 2 Results of the effect of TCS on team performance.

DV	Productivity						Plan Adherence	
	Total		Academic		High-Impact		OLS	
	NB		NB		NB		OLS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Stability	0.436*** (0.092)	4.527*** (0.508)	0.431*** (0.091)	4.511*** (0.509)	0.504** (0.151)	4.681*** (0.575)	0.008 (0.012)	0.240*** (0.034)
Stability ²		−6.780*** (0.948)		−6.770*** (0.950)		−7.029*** (1.061)		−0.385*** (0.062)
Size	0.134*** (0.008)	0.122*** (0.008)	0.128*** (0.008)	0.116*** (0.008)	0.108*** (0.015)	0.092*** (0.015)	0.012*** (0.001)	0.011*** (0.001)
Fund	0.027*** (0.004)	0.023*** (0.004)	0.025*** (0.004)	0.021*** (0.004)	0.040*** (0.002)	0.033** (0.011)	0.000 (0.001)	0.000 (0.001)
Level	−0.150*** (0.021)	−0.138*** (0.021)	−0.136*** (0.021)	−0.125*** (0.021)	−0.011 (0.039)	−0.004 (0.039)	−0.018*** (0.003)	−0.018*** (0.003)
Region	−0.021 (0.013)	−0.018 (0.013)	−0.023 (0.013)	−0.020 (0.013)	0.015 (0.025)	0.015 (0.024)	−0.005** (0.002)	−0.005** (0.002)
Institution	0.061*** (0.011)	0.054*** (0.010)	0.063*** (0.010)	0.057*** (0.010)	0.115*** (0.019)	0.113*** (0.018)	−0.001 (0.001)	−0.001 (0.001)
Student	−0.003 (0.006)	−0.009 (0.006)	−0.002 (0.006)	−0.007 (0.006)	0.006 (0.013)	0.003 (0.013)	−0.001 (0.000)	−0.001 (0.000)
Year	Control							
Discipline	Control							
Sample	4957							
R ²	0.020	0.033	0.021	0.033	0.026	0.030	0.103	0.120

** and *** indicate statistical significance at $p < 0.01$ and $p < 0.001$ respectively. Standard errors are in parentheses.

Reliability assessment of plan adherence. After calculating the plan adherence scores, their reliability was assessed through expert evaluation. Two researchers specializing in management science were invited to evaluate the plan adherence of the project. The researchers scored randomly selected 50 projects based on their expertise and experience, using a scale ranging from 0 to 1, where higher scores denote a greater degree of adherence. Subsequently, we conducted a linear correlation analysis between the expert scores and the adherence scores to test the consistency of these scores. The results indicate a significant positive correlation ($\beta = 0.868$ [0.782, 0.924]***) between the expert scores and the adherence scores calculated in this study, demonstrating that the method proposed effectively measures the project plan adherence.

Effect of TCS on team performance. Table 2 reports the estimation results on the effect of TCS on team total productivity (models 1 and 2), academic productivity (models 3 and 4), high-impact productivity (models 5 and 6), and plan adherence (models 7 and 8). Models 1, 3, 5 and 7 test the linear relationships, while models 2, 4, 6 and 8 test the inverted U-shaped, non-linear relationships between TCS and team performance. Controlling for various confounding factors and introducing the square term of TCS, the multivariable regression models show inverted U-shaped associations between TCS and team performance. TCS is significantly positively related to total productivity ($\beta = 4.527$, $p < 0.001$), academic productivity ($\beta = 4.511$, $p < 0.001$), high-impact productivity ($\beta = 4.681$, $p < 0.001$) and plan adherence ($\beta = 0.240$, $p < 0.001$), and its square term is significantly negatively associated with total productivity ($\beta = -6.780$, $p < 0.001$), academic productivity ($\beta = -6.770$, $p < 0.001$), high-impact productivity ($\beta = -7.029$, $p < 0.001$) and plan adherence ($\beta = -0.385$, $p < 0.001$) (see Column 2, 4, 6, 8 of Table 2), showing the clear inverted U-shaped relationships between TCS and team performance. The results indicate that incorporating more team collaborative stability brings higher performance until stability reaches a turning point, as shown in Fig. 5.

The identification of a U-shaped relationship requires more stringent validation than the mere presence of a statistically significant quadratic term. To rigorously establish the inverted U-shaped relationships between TCS and team performance, we implemented the verification procedure proposed by Lind and Mehlum (2010). Table 3 reports the robustness results for the inverted U-shaped relationships between TCS and team performance. It reveals obvious inverted U-shaped curves with turning points approximating 0.3 within the stability range [0, 1]. The slopes are all significantly positive at low team stability and significantly negative at high team stability. These findings collectively provide robust evidence for the hypothesized inverted U-shaped relationships, indicating that moderate levels of team stability yield optimal performance outcomes.

The impact of leader power. The potential mechanism identified in this study suggests a subtractive relationship between TCS and team performance, where the moderator, i.e., leader power, simultaneously affects knowledge flow in the first mechanism and conflict generation in the second mechanism. Consequently, its overall impact on the inverted U-shaped relationship results in an altered curve shape and a shift in the turning point. To avoid multicollinearity issues, both the moderator ($leaderpower_n$) and the independent variables, including the linear term ($stability_n$) and the quadratic term ($stability_n^2$), were mean-centered before creating interaction terms. Models 1, 3, 5 and 7 include leader power as an explanatory variable to verify the direct effect of leader power. Models 2, 4, 6 and 8 incorporate the interaction terms of leader power with the linear and quadratic terms of stability to test for the moderating effect. The final regression results are presented in Table 4.

The multivariate regression analyses consistently demonstrated a robust positive association between leader power and team performance across all model specifications, with regression coefficients significant at the 0.1% level. Hypothesis 2 was supported.

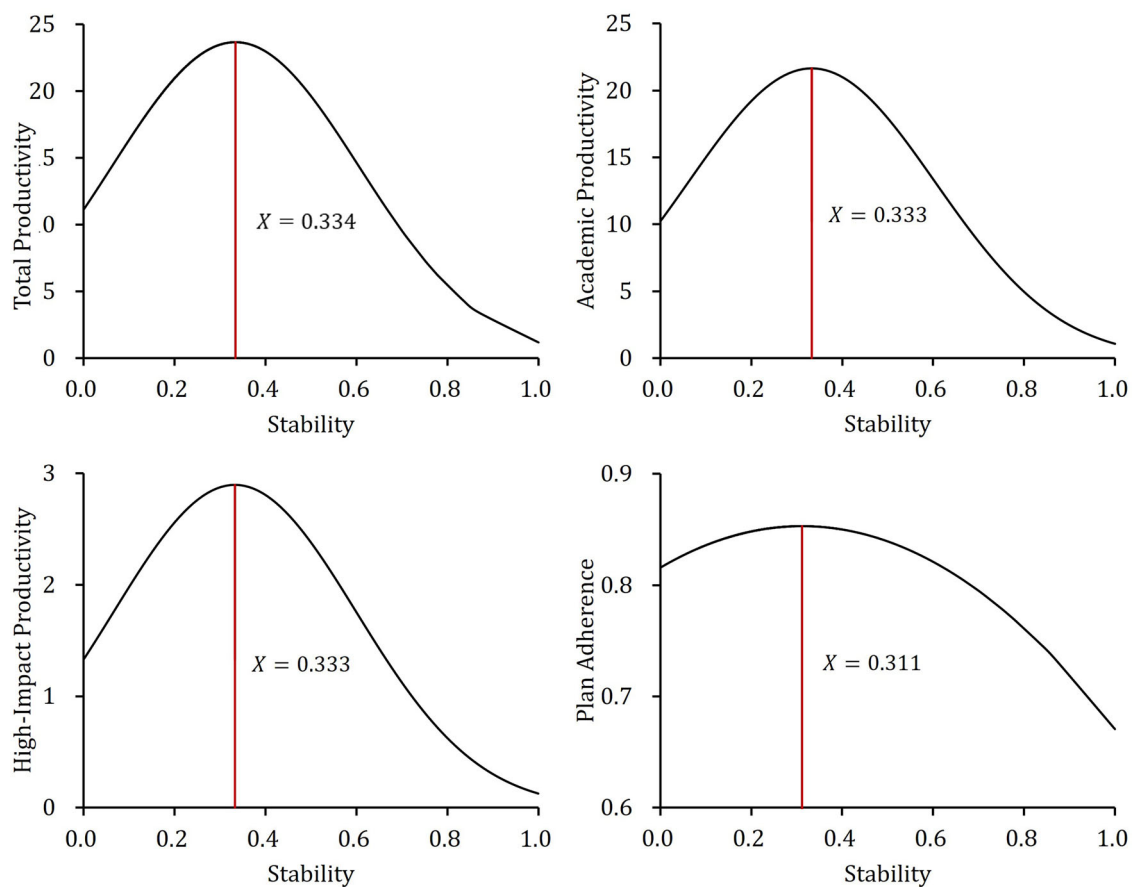


Fig. 5 The inverted U-shaped relationships between TCS and team performance.

Table 3 Robustness test for the inverted U-shaped relationships.				
	Productivity			Plan Adherence
	Total	Academic	High-Impact	
Extreme point	0.334	0.333	0.333	0.311
Lower bound Slop	4.527***	4.511***	4.681***	0.240***
Upper bound Slop	−9.032***	−9.029***	−9.378***	−0.530***
Fieller test (95% confidence interval)	[0.316, 0.363]	[0.315, 0.362]	[0.308, 0.372]	[0.283, 0.349]
Overall test of an Inverted-U shape	$p = 0.000 < 0.001$	$p = 0.000 < 0.001$	$p = 0.000 < 0.001$	$p = 0.000 < 0.001$
*** indicates statistical significance at $p < 0.001$.				

To systematically examine the moderating mechanism, we conducted a two-stage analytical procedure: (1) testing for changes in the curvature of the inverted U-shaped relationship, and (2) assessing shifts in the critical turning point. The moderation analysis revealed statistically significant negative coefficients for the interaction terms of leader power with the quadratic term of stability across all performance metrics: total productivity ($\beta = -6.312, p < 0.001$), academic productivity ($\beta = -6.165, p < 0.001$), high-impact productivity ($\beta = -8.922, p < 0.001$), and plan adherence ($\beta = -0.308, p < 0.001$). These results indicate that enhanced leader power strengthens the positive association between TCS and performance at low to moderate stability levels, and the negative effect is more pronounced at high stability levels. This moderating effect makes the inverted U-shaped relationship between TCS and team performance even steeper.

Further calculation of turning points in the inverted U-shaped relationships reveals a leftward shift under heightened leader

power intensity. As leader power increases from 0 to 2, the team total productivity curve’s turning point shifts from ($x = 0.334$) to ($x = 0.291$), leftward by 0.043; the academic productivity curve’s turning point shifts from ($x = 0.337$) to ($x = 0.292$), leftward by 0.045; the high-impact productivity curve’s turning point shifts from ($x = 0.381$) to ($x = 0.374$), leftward by 0.007; and the team plan adherence curve’s turning point shifts from ($x = 0.353$) to ($x = 0.332$), leftward by 0.021. In summary, Hypothesis 3 was supported, suggesting that increased team leader power causes a leftward shift in the relationship curve between TCS and performance. Figure 6 provides a clearer visualization of the moderating effect of leader power. It can be seen that the inverted U-shaped curve intersects when team stability approaches 1. This indicates that when team stability exceeds a certain threshold, excessive leader power can hinder effective communication and knowledge flow, negatively impacting team performance.

Table 4 Moderating effect of leader power on the inverted U-shaped relationships.

Model	Productivity						Plan Adherence	
	Total		Academic		High-Impact		OLS	
	NB		NB		NB		OLS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Stability	3.110*** (0.595)	4.941*** (0.307)	3.085*** (0.595)	4.814*** (0.311)	2.757*** (0.590)	3.890*** (0.571)	0.137*** (0.034)	0.202*** (0.039)
Stability ²	-3.735** (1.121)	-7.387*** (0.595)	-3.700** (1.121)	-7.146*** (0.604)	-3.012** (1.084)	-5.095*** (1.090)	-0.161** (0.062)	-0.287*** (0.075)
Leaderpower	1.129*** (0.176)	1.253*** (0.193)	1.168*** (0.184)	1.291*** (0.200)	1.568*** (0.152)	1.698*** (0.158)	0.078*** (0.008)	0.082*** (0.008)
Stability × leaderpower		3.351* (1.544)		3.286* (1.556)		6.630*** (1.606)		0.199* (0.082)
Stability ² × leaderpower		-6.312*** (1.659)		-6.165*** (1.621)		-8.922*** (1.795)		-0.308*** (0.082)
Size	0.091*** (0.008)	0.082*** (0.007)	0.085*** (0.008)	0.076*** (0.007)	0.062*** (0.014)	0.057*** (0.014)	0.009*** (0.001)	0.009*** (0.001)
Fund	0.022*** (0.004)	0.018*** (0.004)	0.020*** (0.004)	0.017*** (0.004)	0.029** (0.009)	0.027** (0.009)	0.000 (0.001)	0.000 (0.001)
Level	-0.139*** (0.023)	-0.127*** (0.023)	-0.124*** (0.023)	-0.113*** (0.023)	0.017 (0.038)	0.026 (0.038)	-0.017*** (0.002)	-0.016*** (0.002)
Region	-0.017 (0.013)	-0.019 (0.013)	-0.019 (0.013)	-0.021 (0.013)	0.017 (0.023)	0.016 (0.023)	-0.004** (0.002)	-0.004** (0.002)
Institution	0.049*** (0.011)	0.049*** (0.011)	0.051*** (0.011)	0.051*** (0.011)	0.099*** (0.017)	0.097*** (0.017)	-0.001 (0.001)	-0.002 (0.001)
Student	-0.012 (0.006)	-0.009 (0.006)	-0.009 (0.006)	-0.008 (0.006)	-0.003 (0.012)	-0.003 (0.012)	-0.001 (0.001)	-0.001 (0.001)
Year	Control							
Discipline	Control							
Sample	4957							
R ²	0.054	0.062	0.056	0.063	0.042	0.044	0.156	0.160

*, **, and *** indicate statistical significance at the 5%, 1% and 0.1% levels respectively. Standard errors are in parentheses.

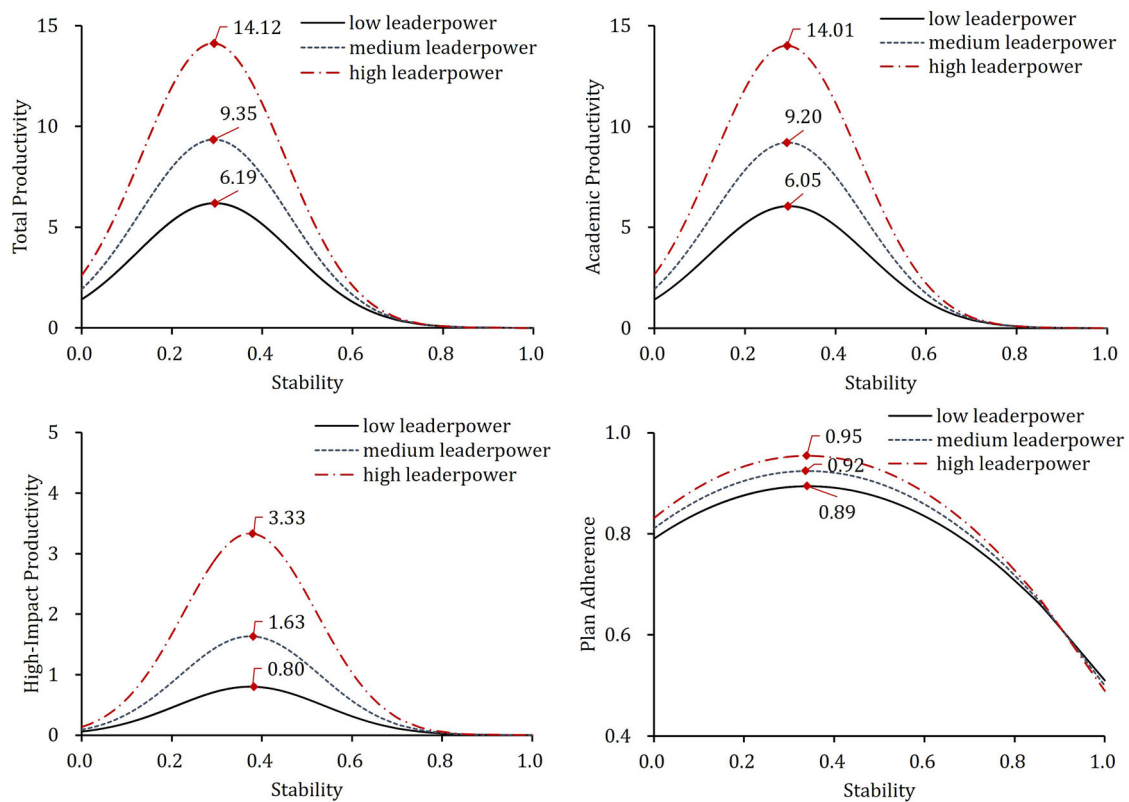


Fig. 6 Moderating effect of leader power on the inverted U-shaped relationships.

Table 5 Robustness test: substitute independent variable.

Model	Productivity						Plan Adherence	
	Total		Academic		High-Impact			
	NB		NB		NB		OLS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Integration	4.410*** (0.155)	3.666*** (0.175)	4.482*** (0.157)	3.710*** (0.180)	4.441*** (0.326)	3.416*** (0.334)	0.277*** (0.021)	0.232*** (0.024)
Integration ²	−4.776*** (0.232)	−3.730*** (0.275)	−4.872*** (0.237)	−3.788*** (0.285)	−4.887*** (0.511)	−3.379*** (0.519)	−0.282*** (0.028)	−0.230*** (0.036)
Leaderpower		1.460*** (0.167)		1.490*** (0.168)		1.669*** (0.120)		0.067*** (0.007)
Integration × leaderpower		6.889*** (0.843)		6.977*** (0.850)		5.810*** (0.993)		0.107 (0.068)
Integration ² × leaderpower		−6.918*** (0.988)		−7.023*** (0.994)		−5.659*** (1.148)		−0.175* (0.082)
Control Variables	Yes							
Sample	6259		6259		6259		6259	

* and *** indicate statistical significance at $p < 0.05$ and $p < 0.001$ respectively. Standard errors are in parentheses.

Additional results and robustness tests. Team stability can also be assessed by the frequency of repeated interactions among team members, providing an alternative perspective for evaluating stability. Here, stability is defined as the strength of repeated collaborative relationships within a team during the execution of NSFC projects. This metric aims to capture the persistence and stability of cooperative relationships among team members. A higher proportion of repeated collaborations indicates that team outputs are more likely derived from stable collaborations among fixed members. Conversely, a lower proportion suggests that the team’s outputs depend more on contributions from transient members who participated only once. Thus, this metric can somewhat reflect team stability from an overall perspective. We substitute the measurement of team stability as Team Collaboration Integration (TCI), which is primarily measured by the proportion of repeated relationships among team members relative to the total number of connections in the network, using the following formula:

$$TCI_i = \frac{N_{repeated\ edges}}{N_{edges}} \tag{6}$$

Since the integration metric quantifies the repeated collaborations from a static perspective without considering member changes over time, data from 2009, 2010, and 2011 can also be included in the regression analysis for robustness testing. This approach increases the sample size to 6,259, allowing for a more comprehensive examination of the relationship between team stability and performance. The final regression results are presented in Table 5. Models 1, 3, 5 and 8 include the linear term ($integration_i$) and the quadratic term ($integration_i^2$) of team integration, while Models 2, 4, 6 and 8 include the moderate variable ($leaderpower_n$) and interaction terms ($integration_i \times leaderpower_n$, $integration_i^2 \times leaderpower_n$).

The quadratic regression models reveal systematic inverted U-shaped relationships between TCI and performance metrics. For the linear terms, TCI demonstrated substantial positive associations with total productivity ($\beta = 4.410, p < 0.001$), academic productivity ($\beta = 4.482, p < 0.001$), high-impact productivity ($\beta = 4.441, p < 0.001$), and plan adherence ($\beta = 0.277, p < 0.001$). In contrast, the quadratic terms exhibited significant negative impact on total productivity ($\beta = -4.776, p < 0.001$), academic productivity ($\beta = -4.872, p < 0.001$), high-impact productivity ($\beta = -4.887, p < 0.001$), and plan adherence ($\beta = -0.282, p < 0.001$). The opposing signs between linear and quadratic terms consistent with

the baseline regression results. The results indicate inverted U-shaped relationships between overall team stability, measured from the perspective of repeated relationships, and team performance. Hypothesis 1 was further supported.

Regarding leader power effects, leader power has significant positive effects across team performance dimensions, including total productivity ($\beta = 1.460, p < 0.001$), academic productivity ($\beta = 1.490, p < 0.001$), high-impact productivity ($\beta = 1.669, p < 0.001$), and plan adherence ($\beta = 0.067, p < 0.001$). As a moderating variable, team leader power strengthens the relationship between TCI and total productivity ($\beta = -6.918, p < 0.001$), academic productivity ($\beta = -7.023, p < 0.001$), high-impact productivity ($\beta = -5.659, p < 0.001$), and plan adherence ($\beta = -0.175, p = 0.033$). These results indicate that superior leader power amplifies performance gains at moderate TCI levels, but accelerates performance decline beyond optimal TCI thresholds, consistent with our hypotheses.

Discussion and conclusion

Scientific teams are crucial drivers of national technological progress and innovation, with team collaboration recognized as a key force in enhancing innovation performance. The NSFC provides competitive funding to scientific teams to pursue breakthrough innovations in science and technology. As the project advances, unexpected innovations may arise, accompanied by new challenges, necessitating real-time adjustments in team composition. NSFC-funded project teams represent one of the largest and most systematically documented cohorts of scientific collaborations in China. The NSFC’s rigorous peer-review system and its role as a primary funder of basic research in China ensure that these teams reflect mainstream collaborative environments within the national scientific ecosystem. This study explores the intrinsic relationship between team collaborative stability and performance in NSFC projects. Although they are distinct performance metrics, we expected collaborative stability to have similar effects on these dimensions. It seeks to identify effective strategies to advance team collaboration and thereby improve performance, finding pathways to enhance innovation performance from the perspective of team collaboration. We draw the following conclusions.

First, this study discovers an inverted U-shaped relationship between team collaborative stability and performance, illustrating an optimal level of stability. When stability is around 0.3, teams achieve peak performance, suggesting that a balanced approach is

beneficial. Excessive stability or instability can be detrimental. This finding aligns with the theories of SAT and TMS, indicating that moderate levels of member stability may introduce fresh ideas while maintaining a cohesive and functional team structure.

Second, this study demonstrates the significant impact of leader power on team performance. Strong leader power positively influences productivity, impact, and plan adherence of NSFC project teams, reflecting the ability of effective leaders to mobilize resources, foster collaboration, and guide the team toward achieving project goals. This supports the theoretical framework of Structural Holes, where leaders act as brokers, enhancing information flow and resource allocation within the team.

Finally, the moderating effect of leader power on the relationship between collaborative stability and performance is particularly noteworthy. Leader power not only enhances overall team performance but also sharpens the inverted U-shaped curve, shifting its turning point to the left. This indicates that leaders with strong power can effectively manage internal resources, thereby preventing knowledge loss and conflicts among members in low-stability teams, ultimately leading to higher performance. Additionally, effective leader power can help mitigate the negative impacts, making it easier to manage high-stability teams by preventing knowledge silos and fostering continuous innovation. However, when stability approaches 1, excessive control by leaders may hinder communication and knowledge flow, negatively impacting team performance. Overall, these findings highlight the crucial role that leader power plays in shaping the relationship between collaborative stability and team performance.

Although causal conclusions cannot be drawn from the current findings, this study provides significant implications for the formation and management of the NSFC project team. By analyzing the internal correlation mechanism between team collaborative stability and performance, more scientific and efficient collaboration patterns can be explored to provide strong support for the sustainable development of scientific teams. This approach enables us to identify the optimal pathways for team performance improvement, which provides practical guidance for project leaders on conducting collaborative NSFC projects. To maximize team performance, leaders should maintain moderate levels of collaborative stability, avoid extreme stability or instability, and leverage their leader power to enhance team integration and resource management. Additionally, project leaders should also balance fostering creativity with maintaining a productive team environment. This study enriches the existing literature by integrating the theories of Structural Adaptation, Transactive Memory Systems, and Structural Holes, offering a comprehensive understanding of the complex dynamics between team stability, leader power, and performance. The dual role of leader power as both a direct enhancer of performance and as a moderator reshaping the performance curve, provides new insights for future research on team dynamics and leader power in scientific research settings.

While this study provides valuable insights, it is essential to acknowledge its limitations. The data is specific to the NSFC Management Science projects, which may restrict the generalizability of the findings. Future research could explore similar dynamics in different contexts or fields, examine the long-term impact of different collaboration strategies on team performance, and investigate additional moderating variables that influence the relationship between collaborative stability and performance. Additionally, our measurement of TCS specifically captures membership dynamics during the funded project period (application to completion), as NSFC teams are institutionally dissolved post-completion. This design reflects the contractual nature of project-based collaboration but acknowledges the limitation in capturing post-project collaborations. Future studies should employ cohort tracking to disentangle project-specific versus

career-accumulated productivity effects. Finally, research impact can be measured through various methods such as the number of patent authorizations (Zhang and Tang, 2018), the citation frequency of papers (Lee et al., 2015), or the journal impact factor (Jing et al., 2023). This study intentionally focuses on journal-tier metrics as one dimension of productivity assessment. Future research can adopt various ways to measure the project impact.

Data availability

The datasets generated and analysed during the current study are available from the corresponding author upon reasonable request.

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Author contributions

Zhizhen Yao conceived and designed the study. Zhizhen Yao and Shuaipu Chen drafted the manuscript. Shuai Zhang and Zhijian Zhang critically revised the manuscript. All authors reviewed the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

Ethical approval was not required as the study did not involve human participants.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Additional information

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